Almost exposed points of real L^1-unit balls are atomic

Run `fa_banach_001`, lane 8

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Status. Candidate partial result, likely valid. This packet does not solve the source paper's full open problem asking whether surjective isometries of the closed unit ball of $L^1[0,1]$ are extremal among nonexpansive self-maps. It gives a sharp obstruction to the almost-exposed-points method used in the source paper and answers the almost-exposed versus exposed comparison inside real $L^1(\mu)$ spaces.

Source. Christian Bargetz, Michael Dymond, Katriin Pirk, "On extremal nonexpansive mappings", arXiv:2409.04292 [1]. In the introduction, the authors note that $L^1[0,1]$ is the only classical Banach space left outside their main theorem and write that they do not know whether surjective isometries of the closed unit ball of $L^1[0,1]$ are extremal. They also define almost exposed points in Definition 3.6 and ask in Remark 3.7 whether every almost exposed point is exposed.

> In condition (i) of Theorem 1.1 the notion of an almost exposed point is weaker than the standard notion of an exposed point, although we do not know whether the two notions may coincide. We postpone further details and discussion on this until Definition 3.6. For now, we note that all Banach spaces with the Radon-Nikodym property satisfy the condition (i) of Theorem 1.1. Consequently, Theorem 1.1 captures all ℓ_p and $L^p[0,1]$ spaces with $1 < p < \infty$. We also capture ℓ_1 (via (i)) and c_0, as well as all $C(K)$ spaces. Since ℓ_∞ and $L_\infty[0,1]$ are isometrically isomorphic to $C(K)$-spaces, the theorem applies to them as well. Hence, arguably the only 'classical' Banach space which evades Theorem 1.1 is $L^1[0,1]$. We do not know whether surjective isometries of the closed unit ball of $L^1[0,1]$ are extremal.

> **Definition 3.6.** Let X be a Banach space and $C \subset X$ a closed and convex set. A point $x \in C$ is called *almost exposed* if the intersection of all hyperplanes supporting C in x is the singleton x.

> *Remark* 3.7. Since every exposed point is almost exposed it seems natural to ask whether the converse is also true, i.e. if every almost exposed point is already an exposed point. We do not know the answer to this question.

In Lemma 3.3, and hence in Proposition 3.2, we are able to show that on sets which are the closed convex hull of its extreme points, the identity is extremal among *linear* nonexpansive mappings. For a corresponding nonlinear result, we need the stronger condition that the underlying convex set is the closed convex hull of its almost exposed points.

Theorem 3.10 (Theorem 1.1, (i))**.** *Let X be a Banach space with the property that its unit ball $\mathbb{B}_X$ is the closed convex hull of its almost exposed points. Then surjective isometries are extremal in the space of nonexpansive self-mappings of the unit ball.*

Proof Intuition

For a unit vector $f \in L^1(\mu)$, every supporting functional of the unit ball at f must equal the sign of f on the set where $f \neq 0$. Outside that set it may vary, but one particular supporting functional, the sign of f on its support and zero elsewhere, already forces any point in all supporting hyperplanes to have norm one, the same sign, and no mass outside the support of f. Thus the whole intersection is not mysterious: it is precisely the simplex of probability densities on the support of f, with the fixed sign of f.

This simplex collapses to a singleton exactly when the support is a measure atom. If the support is not an atom, one can move all mass to a proper positive measure subpiece and get a second point lying in every supporting hyperplane. If the support is an atom, every integrable density on it is constant almost everywhere, so the only point left is f itself.

Statement

Let (Ω, Σ, μ) be a nonzero real measure space and let $\mathbb{B}_1$ denote the closed unit ball of $L^1(\mu)$. A measurable set A with $\mu(A) > 0$ is called an atom if every measurable subset of A has either zero measure or full measure in A.

Theorem 1. *Let $f \in L^1(\mu)$ satisfy $\|f\|_1 = 1$, and put*

$$A = \{\omega : f(\omega) \neq 0\}, \qquad s = \operatorname{sign}(f)$$

with any representative of f. Then the intersection of all hyperplanes supporting $\mathbb{B}_1$ at f, intersected with $\mathbb{B}_1$, is

$$\left\{ sq : q \in L^1(\mu),\ q \geq 0,\ q = 0 \text{ a.e. on } \Omega \setminus A,\ \int_A q \, d\mu = 1 \right\}.$$

Consequently, f is almost exposed in $\mathbb{B}_1$ if and only if A is an atom modulo null sets. In this case f is an exposed point of $\mathbb{B}_1$. Equivalently, the almost exposed points of the real $L^1(\mu)$-unit ball are exactly the exposed points, namely the normalized signed indicators of atoms.

Proof. We first identify the normalized supporting functionals at f. A functional on $L^1(\mu)$ is represented by some $\varphi \in L^\infty(\mu)$. After normalization, a hyperplane supports the unit ball at f precisely when

$$\|\varphi\|_\infty = 1, \qquad \int_\Omega \varphi f \, d\mu = 1.$$

Since $\|f\|_1 = 1$, equality in

$$\int_\Omega \varphi f \, d\mu \le \int_\Omega |\varphi| \, |f| \, d\mu \le \int_\Omega |f| \, d\mu = 1$$

forces $\varphi = s$ almost everywhere on A. Conversely, any $\varphi \in L^\infty(\mu)$ with $\|\varphi\|_\infty = 1$ and $\varphi = s$ a.e. on A is a normalized supporting functional at f.

Now let $g \in \mathbb{B}_1$ lie in every supporting hyperplane at f. In particular, take the supporting functional

$$\varphi_0 = s \, 1_A.$$

Then

$$1 = \int_\Omega \varphi_0 g \, d\mu = \int_A sg \, d\mu \le \int_A |g| \, d\mu \le \|g\|_1 \le 1.$$

All inequalities are equalities. Hence $g = 0$ a.e. on $\Omega \setminus A$, $sg = |g|$ a.e. on A, and $\int_A |g| \, d\mu = 1$. Thus $g = sq$, where $q \ge 0$, $q = 0$ off A, and $\int_A q \, d\mu = 1$.

The converse inclusion is immediate. If $g = sq$ has the displayed properties and φ is any supporting functional at f, then $\varphi = s$ on A, so

$$\int_\Omega \varphi g \, d\mu = \int_A q \, d\mu = 1.$$

Thus g belongs to every supporting hyperplane at f, proving the intersection formula.

It remains to decide when this set is a singleton. If A is an atom, then every real measurable function on A is almost everywhere constant. Hence the only nonnegative q supported in A with $\int_A q \, d\mu = 1$ is $q = \mu(A)^{-1} 1_A$. Since f itself is integrable, nonzero on the atom, and has norm one, it also equals $s\mu(A)^{-1} 1_A$ a.e. Therefore the intersection is $\{f\}$.

If A is not an atom, choose a measurable $B \subset A$ such that

$$0 < \int_B |f| \, d\mu < 1.$$

This is possible because the probability measure $|f| \, d\mu$ on A is equivalent to $\mu|_A$. Define

$$q = \frac{|f| \, 1_B}{\int_B |f| \, d\mu}.$$

Then sq belongs to the displayed intersection, but $sq \ne f$ because $|f| \, d\mu(A \setminus B) > 0$. Hence the intersection is not a singleton. This proves the characterization of almost exposed unit vectors.

Finally, when A is an atom, the functional $s1_A$ exposes f: the same equality argument above shows that any $g \in \mathbb{B}_1$ with $\int_A sg \, d\mu = 1$ must be f. Thus every almost exposed point in real $L^1(\mu)$ is exposed. □

Corollary 1. *The closed unit ball of atomless real $L^1(\mu)$, and in particular of Lebesgue $L^1[0,1]$, has no almost exposed points. Consequently, the hypothesis that the unit ball is the closed convex hull of its almost exposed points fails maximally for $L^1[0,1]$, so Theorem 3.10 of [1] cannot settle the remaining $L^1[0,1]$ extremality problem by this route.*

Proof. Interior points of the unit ball are not contained in any supporting hyperplane of the ball. Boundary points have norm one, and by the theorem an almost exposed boundary point would have atomic support. Atomless measure spaces have no such support sets. □

Verification and limitations

The proof is self-contained and uses only the standard duality $(L^1(\mu))^* = L^\infty(\mu)$ in the regime where supporting functionals are represented by L^∞-functions. No computational checker is involved.

The result is intentionally stated for real $L^1(\mu)$. A complex version should be formulated separately, since supporting hyperplanes are real affine hyperplanes and the signs become unimodular phase functions.

This packet does not prove or disprove that surjective isometries of the closed unit ball of $L^1[0,1]$ are extremal among nonexpansive self-maps. It does show that the almost-exposed convex-hull mechanism in [1] gives no traction in atomless L^1, because there are no almost exposed points to start from.

Novelty check

The local run indexes were searched for `2409.04292`, `extremal nonexpansive`, `almost exposed`, and `L^1[0,1]`, with no exact duplicate packet found. A bounded web search on 2026-06-19 for exact phrases including "almost exposed L^1 unit ball", "almost exposed point L1 Banach", and the source paper's $L^1[0,1]$ extremality language did not reveal a later answer. This supports but does not prove novelty; human review is requested.

References

[1] Christian Bargetz, Michael Dymond, and Katriin Pirk. On extremal nonexpansive mappings. arXiv:2409.04292, 2024. `https://arxiv.org/abs/2409.04292`.